OPERATION Department
SIMHADRI

Note No :1

Ref:SMTPP/OPERATION/2024-25/OPR_LMI/693913 **Date:**22/11/2024(15:48:41)

Subject:LMI on Safe Operation and Maintenance of Pulverised Fuel Plant

Site LMI on Safe Operation and Maintenance of Pulverised Fuel Plant (LMI Tag: LMI/OD/OPS/SYST/009) revised under revision no. 7 .

Please review.

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LMI is in line with the OD.

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NTPC **SIMHADRI**

LOCATION MANAGEMENT INSTRUCTION

The Safe Operation and Maintenance of Pulverised Fuel Plant

LMI/OD/OPS/SYST/009 Rev: 07
NOVEMBER-2024

Approved for Implementation by HOP (SIMHADRI)

LMI APPROVAL DOCUMENT

1	**LMI NAME**	The Safe Operation and Maintenance of Pulverised Fuel Plant
2	**LMI No.**	LMI/OD/OPS/SYST/009 Rev:07 November-2024
3	**Reference Document**	COS/ISO/00/OD/OPS/SYST/009 September-2024
4	**Superseded Document**	LMI/OD/OPS/SYST/009 REV:06 October-2022
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TITLE	The Safe Operation and Maintenance of Pulverised Fuel Plant
LMI No.	LMI/OD/OPS/SYST/009 Rev:07

1.0 INTRODUCTION:

Simhadri is provided with medium speed vertical spindle pressurized pulverisers (XRP 1003- bowl mill) for pulverizing and supplying coal to furnace for direct firing. Coal is fed to the center of the pulveriser from a variable speed feeder which in turn is fed by raw coal bunkers. The coal passes between the bull ring and grinding rolls which impart the necessary pressure for grinding. Tempered hot primary air enters the mill housing from side and carries the pulverized coal to furnace through classifier, wherein coal of required fineness is only carried to furnace and rest is returned to mill for further grinding. Any tramp iron or difficult to grind foreign material in coal is rejected and collected through scrappers, pyrite hoppers and pneumatic conveying system.

The ignition of finely divided pulverized fuel in suspension in air can cause significant explosion and emission of flame. As detailed above these suspensions are essential parts of the processes inside PF mill systems, boiler furnaces and in view of some unburnt carried, in precipitator hoppers.

The objective of this LMI is to detail operation and maintenance practices to be adopted for the ensuring safety of pulverized fuel plant under different operating conditions.

2.0 SCOPE:

The scope of this LMI is to detail the methods to be adopted for the safe operation and maintenance of pulverized fuel plant. Pulverized fuel plant refers to raw coal bunkers, raw coal feeders, mills and associated piping, air ducting from the point where hot and tempering air ducts leave their respective main supply ducts.

3.0 REFERENCE DOCUMENT:

Operation directive COS-ISO-00-OD/OPS/SYST/009 circulated by operation services - NTPC Corporate Center dated 03/08/2022.

4.0 SUPERSEDED DOCUMENTS:

LMI/OD/OPS/SYST/009/, ISSUE NO.1; REVISION NO. 6.00; DATE: Oct'2022

4.1 REVISION

LMI format changed as per new guidelines.

5.0 SPECIFIC INSTRUCTIONS:

The objectives of the LMI are sought to be achieved through

1. Preventing concentration of combustibles in suspension in air, falling within the explosive concentration limits

2. Preventing source of ignition of adequate energy

3. Preventing pulverized fuel deposits

4. Establishing and educating personnel regarding emergency operation procedures

6.0 METHODOLOGY OF IMPLEMENTATION & RESPONSIBILITY CENTERS:

6.1. Preventing concentration of combustibles in suspension in air:

A) During start up and shut down of mill, development of explosive mixtures is common. The air fuel ratio's change from very weak to normal as mill is bought into service. Explosions cannot be sustained in mixtures where the air fuel ratio is weaker than 5:1. The explosive effect of air fuel mixtures during normal operation and start- ups/shutdowns can be mitigated by adopting the following operation and maintenance practices.

6.1.1 During Planned Start-up of a Mill:

1. Establish ignition energy in the boiler for the mill to be started.

2. Ensure mill reject chamber is clear

3. Ensure mill outlet temperature is less than 50ºC

4. Open all burner isolation gates & mill discharge valve of the mill to be started. Open Cold air gate and check that all mill start permissive are available.

5. Hot air gate, hot air damper & cold air damper should remain closed.

6. Start the mill and establish air flow to remove all coal through mill reject (cold air damper opens on auto as per logic)

7. Open hot air gate and modulate hot air damper/cold air damper to maintain air flow and mill outlet temperature. When the mill current comes down, start the feeder.

8. Maintain the parameters as per normal procedure

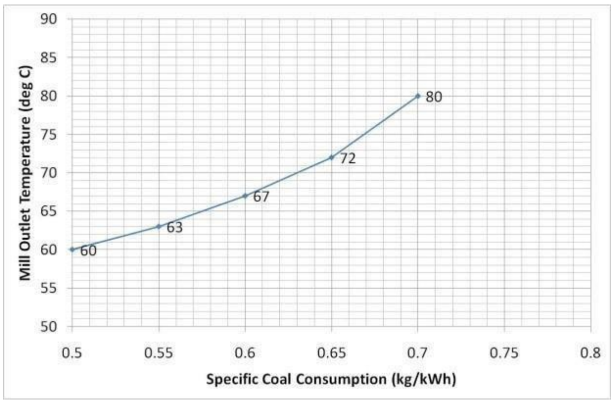
9. Do not start a mill if there is evidence of fire in that system

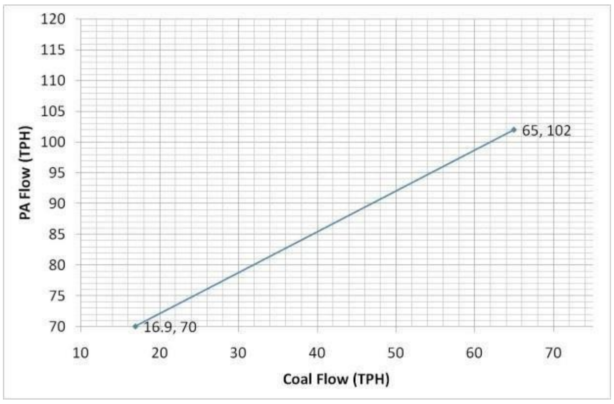
10. During Mill taking in Service / out of service Ignition permissive to be checked or oil gun to be taken as per OEM recommendation.

6.1.2 During Normal Operation of the Mill:

1. Limiting value of Mill outlet temperature to be further reduced in case of imported Coal as per SOP on imported Coal as below:

Specific Coal Consumption (ton/MWh)	Mill O/L Temp (°C)
0.5	60
0.55	63
0.60	65
0.65	70
0.70	80-85





2. In case of Pellet Firing Mill inlet temperature set point should be controlled in such a way that operator cannot give more than 180 °C set point & Alarm for mill outlet temperature >70°C to be provided.

3. Trend of Mill outlet temperature to be monitored. If it is rising abnormally, it may be an indication of Mill fire. If so, Immediately Mill is to be stopped and emergency operation for Mill fire to be followed.

4. Mill outlet temperature high alarm to be maintained in good working condition (Mill outlet temp. high alarm set point may be set at 80°C). Its thermocouple should be checked periodically by C&I. the same alarm should be checked jointly by C&I and Operation once in every six months. Rate of rise of temp alarm also to be provided.

5. In the event of pulveriser outlet temp going high (>90 °C), temp control is released from Auto mode. The hot air gate and hot air damper closes and cold air damper open to 60%. These interlocks should be checked during annual overhaul. Any malfunctioning of the above interlocks observed during operation is to be reported and rectified.

6. While using blended coal, care is taken to ensure that Pulverized coal & air mixture is not lean in order to avoid mill fire.

7. Pulverised coal pipe temperature monitoring to be done to observe any choking tendency in the mill

8. Pulveriser inlet PA temperature & pressure to be monitored during mill running. If there is sharp rise or fluctuation in PA Inlet pressure to mill, it may be an indication of mill fire. If so, immediately mill is to be stopped and Emergency operation for mill fire to be followed.

9. Local operator should take rounds periodically around the mill and auxiliaries and report about any coal dust leakage, oil leakage, accumulation of coal dust, smoke, sparkles coming out etc.

10. Instruments for indication of Mill DP should be maintained in good working condition.

11. Mill reject evacuation system should operate in auto. Reject should not be allowed to accumulate for long in reject chamber. Mill tramp iron discharge gate should normally be in open condition to allow the free discharge of foreign materials in the pyrites collections system.

12. Accumulated debris like paper, wood, straw etc. if any inside mill to be removed when mill is opened for inspection.

6.1.3 Planned Shutdown of a Mill:

1. Reduce the feeder speed of the mill to be taken out of service & adjust that in the other mills so that the total coal flow remains constant.

2. Reduce the mill outlet temperature slowly to 55 °C

3. Slowly bring the feeder speed to minimum. (13 to 15 T/hr)

4. Continue feeding coal till mill outlet temperatures reach 55°C as mill trip is provided at 50°C. Then stop feeder.

5. Stop the feeder manually. HAG closes in Auto. Ensure full closure of HAD.

6. Make the mill empty by giving cold air through CAD. (Ensure PA Flow of 70T/hr)

7. Mill should be run for around 10-15 minutes after feeder is taken out of service.

8. Ensure mill current at no-load value.

9. Reduce the mill outlet temperature to around 50 °C by giving cold air.

10. Reduce cold air gradually & close CAD fully. Stop the pulveriser.

B) An explosive mixture can also result during operation caused by:

1. loss of coal feed
2. faulty control or indication systems
3. low air flows
4. lean coal-air mixtures and high temperatures arising from manual controls.

B. 1) In case there is a loss of coal feed partially or fully as against the set and indicated coal flows this would get indicated by increase in mill outlet temperatures. Do not allow mill temperatures to exceed 80°C. In such conditions the following procedure is to be adopted.

i) Establish appropriate oil burners for the mill concerned.

ii) Put mill controls on manual, reduce primary air flow to minimum value and adjust coal feeder output as necessary.

iii) Open cold air damper to control mill outlet temperatures.

iv) Try to establish coal flow.

v) If the coal flow cannot be restored, shut down the mill.

vi) During the above; do not open any part of milling system to atmosphere like feeder doors.

Responsibility: UCE

To establish coal flow by hammering on bunker/ feeder body and not opening doors:

Responsibility: Sec. Head of Mech. Maint.

B. 2) Excessive mill outlet temperatures for reasons other than loss of coal feed such as improper hot and cold air damper modulation may also result in fires. Do not allow pulveriser temperatures to exceed 80 °C by taking controls in manual if necessary. There is a protection of HAG closing with mill temperatures exceeding 94 °C.

Responsibility: UCE

B. 3) Air transport velocities > 20m/sec will minimize deposition of pulverized fuel in coal pipes.

i) To sustain air velocities greater than this, air flow > 70 T/hr is always to be maintained in the system Responsibility – UCE

ii) Provision of alarm for air flows less than 70 T/hr - Responsibility - Head of C&I

iii) Close monitoring of coal pipe temperatures just before BIG to be done during normal running of the pulveriser. Mill outlet temp higher than coal pipe temp by 3 to 5 °C indicates healthy operating condition. Any deviations such as mill outlet temp remains lower than PF pipe temp, wide variations in temperatures of four corner PF pipes of a mill must be highlighted and investigated. Accuracy of the temperature indications must be checked & rectified immediately.

Responsibility – UCE

iv) Dirty air flow test is to be conducted twice in a year to map air and coal flows in each pipe of a particular mill and to determine pipe with choking tendencies.

Responsibility – Head of EEMG

v) Removal of choking as indicated by tests and replacement of worn out orifices during overhaul/ when noticed depending requirement in case of imbalance of air flow.

Responsibility – Head of Mech. Maint

B. 4) The accumulation of an explosive mixture in the furnace is an event that occurs when there is loss ignition during operation. Loss of ignition protection is provided to stop the pulveriser and coal feed in case of such eventuality.

B. 5) There is a possibility of fire in pulverized fuel plant on account of faulty indications and controls because of component failure of furnace supervisory and safeguard system. It is to be ensured that failure of components of FSSS system or loss of motive power to these components or to field equipment controlled by these components does not lead to unsafe process conditions without reliance on the skill of operator. Integrity of the system is to be checked and confirmed after every overhaul

Responsibility - Head of C&I

6.2. Preventing sources of ignition of adequate energy:

6.2.1 Furnace over pressurizations can result in ignition sources being introduced into the milling systems and can result in temporary reduction of PF transport velocities and consequent deposition/reverse flow in running mills. Failure of primary air supply to the mill with mill open to furnace and continuing to pulverize coal can also result in ignition sources being introduced into PF plant. In case of furnace pressures going towards positive, operator has to take necessary action if required through manual controls for preventing the same.

Responsibility- UCE

6.2.2 Welding and Cutting activities near to PF dust deposits and dust clouds can also be a source of ignition. Strict compliance to Safety / PTW norms to be adhered to.

Responsibility – Head of Mech. Maint

6.3 Preventing Pulverized Fuel deposits:

6.3.1 There could be accumulation of coal deposits in the mill scrapper chamber because

a) Excessive clearance than specified between scrapper and mill bottom plate

b) Blockage of mill reject passage

i) Free passage of mill reject system to be checked manually once in four hours.

Responsibility- UCE

ii) Any Alarm condition in reject handling system to be immediately intimated to Concerned maintenance group for rectification. During the down time of auto reject evacuation system manual reject evacuation system is to be operated Periodically.

Responsibility – UCE

iii) Chokage in reject handling system to be cleared immediately. If the same is not possible to be done on-line, mill shut down to be taken.

Responsibility- Head of Mech. Maint.

iv) Clearance between scrapper and bottom plate to be checked and maintained to 8-10mm during preventive maintenance.

Responsibility- Head of Mech. Maint.

6.3.2 Accumulation of pulverized fuel in areas above bowl or on coal pipes on account of leakages and worn out liners/ pipelines to be avoided as there could be spontaneous combustion of accumulated coal even where air is at ambient temperatures. Preventive maintenance action to include strengthening of all eroded portions to eliminate leakage during mill running conditions. PM activities to also include checking chokage in classifier. Walk downs to be conducted once in every day to identify leakages/ accumulations. All identified/ reported leakages to be attended immediately, if possible, on-line.

Responsibility- Head of Mech. Maint

6.3.3 Coal accumulations as a result of leakages to be cleared immediately after attending leakages. Housekeeping of areas above classifier, PA ducts, feeder body and coal pipes to be done by default once in two days.

Responsibility- Housekeeping In-charge.

6.3.4 Spontaneous combustion of stagnant coal in bunker can also cause fire. Bunkers associated with mills which are not in service for long periods are most susceptible to these fires.

a) During unit running condition no mill is to be kept idle for long periods with filled bunker. Before long shutdowns bunkers are to be emptied as for as possible within safe operation limits.

Responsibility- UCE.

b) Despite the presence of seal air to feeder there can always be small primary air flow from the mill to bunker which increases if the bunker coal level is low during startup of mill. Feeder discharge valve of standby mill should be always be in closed condition. Bunker o/l gates of empty bunkers be kept closed.

Responsibility- UCE.

c) If there is a suspected fire in bunker, primary aim to be to empty the bunker directly into lorries for disposal. During these hot spots on bunker & feeder to be sprayed with water. In case the bunker is to be emptied through mill, Mill Mtc. group should continuously monitor the fire. Non- essential personnel should be withdrawn from the mill area. After emptying the bunker, the insides of the bunker to be cleaned of any hot spots by water jets. Spray water supply to feeder to be provided by Mech Mtc.

Responsibility- Head of Operation & Head of Mech. Maint.

6.4 Emergency operation procedures:

6.4.1 Emergency procedures during fire in pulverized fuel plant The most common indications of fire in milling system are:

a) A high and rapidly increasing mill outlet temperature without any other cause

b) Paint peeling from mill and other piping.

c) Visible signs of overheating, smoke emission or sparks

6.4.2 If a mill system fire occurs the following procedure is to be followed:

Stopping of Mill by Emergency Shutdown (FIRE):

1. In case of sharp rise of mill outlet temperature/indication of mill fire like paint peel off in PC pipes or mill body, Mill should be hand tripped immediately. Tripping of mill trips the feeder. Ensure the same.

2. Ensure closing of HAG, CAG, HAD & CAD.

3. Assure that flow path exists to the furnace for steam inerting (MDVs & BIGs shall remain open and ID/FD fans shall be running)

4. Open the pulveriser steam inerting valve to supply the required steam flow to the mill, to achieve an inert atmosphere

5. Steam inerting to be done for a minimum duration of 5 minutes.

6. Allow the mill to be cooled down till mill outlet temperature is less than 50°C

7. Restart the mill without air to empty the mill through reject system.

8. Issue PTW on mill for internal inspection

Responsibility- UCE.

6.4.3 The following recommendations on Steam inerting system by the committee to be implemented.

1. Existing manual valves to be replaced with motorized valves for individual mill steam inerting valves as well as main isolation valve system of all units.

Responsibility- Head of Mech Maint / Head of EMD

2. Steam trap are to be installed in mill inerting system in stage-1 units in-line with stage-2 units. Responsibility- Head Mech Maint

3. Two No's Pressure transmitters to be installed in stage-1 units for remote indication of inerting steam pressure.

Responsibility: Head of C&I

6.4.4 Issues to be taken up with Engg & BHEL

a. Auto operation philosophy of mill inerting system to be incorporated along with relevant modification in FSSS Logics.

b. Tripping of mill in case of feeder trip

c. Multiple mill inerting in case of MFT/Runback. One mill on either side needs to be inerted simultaneously, till all tripped mills inerting is completed.

Responsibility- CC- OS

6.4.5 Starting of Mill after Emergency Shutdown (FIRE):

1. After clearance from Mill maintenance group, start the mill & ensure complete emptiness of the mill.

2. Purge the PC Pipes one after another with cold air for 15-20 minutes.

3. Hot air flow to mill to be given slowly for raising the mill outlet temperature to requisite value (65 °C). Establish air flow as per mill air flow curve (i.e. 70 T/hr)

4. Start feeder & maintain air flow, mill outlet Temp as per curves.

5. Load the mill as per requirement & keep close watch on mill current, PA flow, bowl DP, inlet PA temperature, mill outlet temperature, PC pipe temperatures & rejects etc.

Responsibility- UCE

6.4.6 Proper Purging of Mill:

Any mill which has tripped on load or which has not been emptied before stopping must be purged as follows, before being taken back in service:

1. Restart the mill with CAG & HAG closed & purge out accumulated coal from mill bowl through mill reject.

2. After mill bowl is emptied, mill outlet PF pipes have to be purged only with regulated cold air flow. Ensure that mill outlet temp & PF pipe temps are low (<50 °C) before admitting any air and also ensure that these temps don't show a rising trend as soon as Cold air is admitted. If the temperature shows a rising trend, cut off airflow & allow the mill/PF pipes to cool down further if required, BIGs may be closed to cutoff air flow completely & allow natural cooling.

3. For taking the mill back in service after purging, HAG & HAD are to be opened & Hot Air admitted only after the above temperature stabilize below 60 °C.

4. No attempt is to be made to take back a mill handling blended/imported coal immediately after it has tripped on load.

Responsibility- UCE.

6.4.7 Education regarding emergency procedures:

Concerned executives should learn by studying LMI's and manuals.

7. EXCEPTION REPORTING:

Any incidence of fire in the milling system is to be recorded in the following format.

Responsibility – SCE

Description of fire incidence	Unit No.	Location of fire	Actions taken in respective shift	Remarks

8. RESPONSIBILITIES:

The respective responsible departments in carrying out different tasks in relation to safe operation and maintenance of pulverized fuel plant as detailed in the LMI shall maintain their records updated

9. RECORD KEEPING:

The records to be maintained under this LMI include

a) Exception reporting on fire incidence.

10. REVIEW:

This document shall be reviewed by Head of Project on two-yearly basis.

11. DISTRIBUTION LIST: LAN

ANNEXURE-1

PRECAUTIONS DURING BIOMASS PELLETS CO-FIRING

1.0	MILL OPERATING PARAMETERS AND OPERATION PROCEDURE
	• Maintain Mill outlet temperature 60 °C • Mill inlet temperature not more than 190 °C • Observe all 4-coal pipe temperature of Mills with biomass regularly. In case of any deviation from rest of the pipes (any abrupt rise or fall), stop the mill and give PTW for inspection. • Prestart and post shutdown purging of mill is mandatory. • Continuous monitoring of mill parameters most importantly mills inlet air temperature, mill outlet temperature, mill current and mill DP. • Mill inlet temperature should not be more than 190 °C in any case • Mill reject system to be kept under observation continuously • Operator should be deployed at local for monitoring in wake of fire hazard. • In case of fire, mill should be tripped and steam inerting should be done immediately by opening steam inerting motorized valve, keeping watch on furnace pressure. Fire tender may also be used if required. • Clinkering and slagging tendency to be observed from local as well as from rise in SH zone FG temperature. Soot blowing and LRSB to be done accordingly

2.0	CHANGE OVER TO BIOMASS MODE AND VICE-VERSA
	For operator ease Biomass On/OFF buttons have been provided on individual Mill screen. Whenever Biomass ON is selected on screen then following changes happen in the Mill logics-: 1. Mill temperature control by CAD (cold air damper) of the mill shifts to Mill inlet temperature control as per the Inlet temperature set point. 2. Mill HAG (hot air Gate) will close on protection if Mill inlet temperature reaches 195 deg. Alarm for Mill inlet temperature is provided at 185 deg. 3. HAG (hot air gate) protection at mill outlet temperature 95 °C will also remain in service irrespective of Biomass ON/OFF mode **<u>Changeover to Biomass mode ON mode:</u>** 1. Put the CAD& HAD control to manual mode. 2. Ensure the set value for Mill inlet air temperature to 180-190 deg. 3. Maintain the position of HAD & CAD so as to bring the actual Mill inlet temperature in the set value range. Wait for temperature stabilization 4. Select Biomass ON button. 5. Put CAD & HAD in auto. 6. Mill CAD control will shift to Mill inlet temperature. **<u>Changeover to Biomass OFF mode</u>**: 1. Put the CAD& HAD control to manual mode. 2. Bring the set value of Mill temperature near to actual running Mill outlet temperature by manually adjusting the Mill HAD/CAD. 3. Select Biomass OFF mode, CAD control will shift to Mill outlet temperature. 4. Put CAD & HAD in auto.
3.0	ACTIONS TO BE TAKEN IN CASE OF FIRE DURING BIOMASS FIRING
	Do's • Call the CISF fire personnel for immediate assistance. • Increase the feeder speed to maximum and dump the coal to mill for few minutes. • Stop the feeder after few minutes. • Close HAG from control room and CAG & Mill seal air valve from local. • If the pulveriser temperature continues to rise, admit inert steam by opening the motorized steam inerting valve from control room. • If fire occurs in the feeder then, close the feeder seal air valve and open the hydrant valve provided near individual feeder.

Don'ts

- Do not admit cold primary air in order to cool the mill.
- Do not expose any part of the mill (roller, scrapper chamber and tramp iron gate for any inspection until the mill temp comes down to normal value).